

Amazon Product Analysis

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Abstract

Amazon is a multinational e-commerce company and has experienced a 41% decline in product purchases during its Prime Day event in 2025 compared to the previous year. As a result, both the *Marketing* and *Sales* departments at Amazon are looking to improve their customer experience. They seek the help of the Lead Data Analyst to provide them with a comprehensive product report which will examine the products carried by Amazon. The *Marketing* department will use this information to deliver product-specific digital promotion while the *Sales* department will use this information to suggest a range of products most suitable to each customer.

Research Questions

A product analysis will be performed for the marketing and sales departments with a focus on answering the following research questions.

Question 1

How does the price, rating and discount compare for products belonging to different categories?

Purpose

Obtain an overview of the products sold by Amazon to make it easier to implement guided marketing and sales strategies.

Impact Hypothesis

When a sales employee is making efforts to sell a product to a future customer, the employee will have sufficient knowledge to answer any questions about the product, including questions relating to pricing and potential discounts.

Question 2

How can the product characteristics be used to predict the product category?

Purpose

Classify new products to ensure appropriate marketing and sales strategies are applied for market success.

Impact Hypothesis

A member of the *Marketing* team is asked to curate digital content to promote a new product. In this case product category classifier models can be used to identify the already existing digital materials prepared for products belonging to the same category.

Question 3

How can product descriptions be used to identify similar products?

Purpose

Match new products to existing products for segmentation and recommendation strategies.

Impact Hypothesis

A sales employee needs to find products similar to the one described by a customer using the product information retrieval models.

Methodology

Preliminary Analysis

Extract descriptive statistics such as means and frequencies. Compute correlations to observe potential relationships between variables. Visualize variable distributions to observe general trends and outliers using boxplots and histograms.

Classification Models

Based on the results of the preliminary analysis, construct three different product category classifiers: a linear multinomial logistic regression model, and two non-linear classifiers: a k-nearest neighbours model and a decision tree model. Compare the three models using evaluation metrics such as accuracy, precision and recall.

Information Retrieval Models

Construct multiple unigram and bigram vector space models to process product descriptions. Apply different text preprocessing techniques (stopword removal, lemmatization, etc.) and use both the TF and TD-IDF statistical weight measures to evaluate importance of words. Calculate accuracy, precision and recall to evaluate search relevance.

Data Analysis

The *Amazon Sales Dataset* will be the primary source of data for this project which contains over 1000 authentic Amazon product records scraped using Python. The dataset contains 16 variables, and *Figure 1* provides an overview of the relevant characteristics of the dataset.

Figure 1: Overview of Variables

Name	Description	Type
product_name	String containing product name	Text

category	Pipe separated string containing product categories	Text
discounted_price	Discounted price of product in Indian Rupees (INR)	Numerical with units
actual_price	Price of product in INR	Numerical with units
discount_percentage	Percentage discount on product	Numerical with units
rating	Rating of product from 1 to 5	Ordinal, categorical
rating_count	Number of individuals who voted for a product rating	Numerical
about_product	Description of product	Text

Figure 2: Overview of Data

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories Accessories&Peripherals ...	₹399	₹1,099	64%	4.2	24,269	High Compatibility : Compatible With iPhone 12...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories Accessories&Peripherals ...	₹199	₹349	43%	4.0	43,994	Compatible with all Type C enabled devices, be...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories Accessories&Peripherals ...	₹199	₹1,899	90%	3.9	7,928	【 Fast Charger& Data Sync】 -With built-in safet...
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories Accessories&Peripherals ...	₹329	₹699	53%	4.2	94,363	The boAt Deuce USB 300 2 in 1 cable is compati...
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories Accessories&Peripherals ...	₹154	₹399	61%	4.2	16,905	[CHARGE & SYNC FUNCTION]- This cable comes wit...

Figure 3: Data Types and Missing Values

#	Column	Non-Null Count	Dtype
0	product_id	1465 non-null	object
1	product_name	1465 non-null	object
2	category	1465 non-null	object
3	discounted_price	1465 non-null	object
4	actual_price	1465 non-null	object
5	discount_percentage	1465 non-null	object
6	rating	1465 non-null	object
7	rating_count	1463 non-null	object
8	about_product	1465 non-null	object
9	user_id	1465 non-null	object
10	user_name	1465 non-null	object
11	review_id	1465 non-null	object
12	review_title	1465 non-null	object
13	review_content	1465 non-null	object
14	img_link	1465 non-null	object
15	product_link	1465 non-null	object

The dataset contains no missing values and the quantitative variables will need to be converted to their numerical equivalent data types.

Data Cleaning

Characters and punctuation will be removed from the numerical variables before converting them to their float equivalents.

Figure 4: Cleaned Dataset

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories Accessories&Peripherals ...	399.0	1099.0	64.0	4.2	24269.0	High Compatibility : Compatible With iPhone 12...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories Accessories&Peripherals ...	199.0	349.0	43.0	4.0	43994.0	Compatible with all Type C enabled devices, be...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories Accessories&Peripherals ...	199.0	1899.0	90.0	3.9	7928.0	【 Fast Charger& Data Sync】 -With built-in safet...
3	B08HJDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories Accessories&Peripherals ...	329.0	699.0	53.0	4.2	94363.0	The boAt Deuce USB 300 2 in 1 cable is compati...
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories Accessories&Peripherals ...	154.0	399.0	61.0	4.2	16905.0	[CHARGE & SYNC FUNCTION]- This cable comes wit...

Summary of Variables

Figure 5: Summary of Quantitative Variables

	actual_price	discounted_price	discount_percentage	rating_count
count	1465.000000	1465.000000	1465.000000	1463.000000
mean	5444.990635	3125.310874	47.691468	18295.541353
std	10874.826864	6944.304394	21.635905	42753.864952
min	39.000000	39.000000	0.000000	2.000000
25%	800.000000	325.000000	32.000000	1186.000000
50%	1650.000000	799.000000	50.000000	5179.000000
75%	4295.000000	1999.000000	63.000000	17336.500000
max	139900.000000	77990.000000	94.000000	426973.000000

The price of products ranges from ₹39 to ₹139,900 with the average price being about ₹5445. This suggests that there are extreme outliers present in both the actual and discounted price variables.

Figure 6: Overview of Product Categories

	count
category	
Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables	233
Electronics WearableTechnology SmartWatches	76
Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones	68
Electronics HomeTheater,TV&Video Televisions SmartTelevisions	63
Electronics Headphones,Earbuds&Accessories Headphones In-Ear	52
...	...
Home&Kitchen Kitchen&HomeAppliances SmallKitchenAppliances RotiMakers	1
Home&Kitchen Heating,Cooling&AirQuality Parts&Accessories FanParts&Accessories	1
Home&Kitchen Kitchen&HomeAppliances SmallKitchenAppliances StandMixers	1
Home&Kitchen Heating,Cooling&AirQuality Fans PedestalFans	1
Home&Kitchen Kitchen&HomeAppliances Vacuum,Cleaning&Ironing Vacuums&FloorCare VacuumAccessories VacuumBags HandheldBags	1

211 rows x 1 columns

There are 211 product categories 205 of which correspond to only one product. To address this massive data imbalance, only the categories with 30 or more products will be used for the analysis.

Figure 7: Product Categories with 30 or More Products

category	count
Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables	227
Electronics Wearable Technology SmartWatches	70
Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones	68
Electronics Home Theater, TV&Video Televisions SmartTelevisions	61
Electronics Headphones, Earbuds&Accessories Headphones In-Ear	47
Electronics Home Theater, TV&Video Accessories RemoteControls	39

Figure 8: Overview of Product Ratings

rating	count
4.1	244
4.3	230
4.2	228
4.0	129
3.9	123
4.4	123
3.8	86
4.5	75
4	52
3.7	42
3.6	35
3.5	26
4.6	17
3.3	16

Although Figure 8 does not display all possible values of ratings, several correspond to fewer than 30 products. Similarly to product categories, only the ratings which correspond to greater than 30 categories will be considered as per the law of large numbers.

Figure 9: Product Ratings with 30 or More Products

rating	count
4.1	102
4.2	101
4.3	94
4.0	70
3.9	54

Note that the counts differ as these are the ratings with 30 or more products after filtering the dataset to only include categories with 30 or more products.

Measures of Central Tendency

Figure 10: Averages by Product Category

category	actual_price	discounted_price	discount_percentage	rating_count
Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables	824.469787	305.710745	60.579787	15163.984043
Electronics Headphones,Earbuds&Accessories Headphones In-Ear	2427.437500	932.250000	53.062500	111749.125000
Electronics Home Theater,TV&Video Accessories RemoteControls	1126.833333	386.666667	57.277778	848.166667
Electronics Home Theater,TV&Video Televisions SmartTelevisions	38955.169492	24383.966102	37.627119	12739.000000
Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones	20613.138462	15773.938462	23.092308	37995.200000
Electronics Wearable Technology SmartWatches	9181.762712	2429.423729	69.864407	26073.508475

Smart televisions have the highest average price compared to all other categories while smart watches have the highest offered average discount.

Figure 11: Averages by Product Rating

rating	actual_price	discounted_price	discount_percentage	rating_count
3.9	5900.542963	3417.106296	61.111111	15387.814815
4.0	4193.957143	2394.284143	54.628571	16884.257143
4.1	9970.901961	6498.057745	46.019608	54020.588235
4.2	11581.019802	6349.376238	54.495050	26996.485149
4.3	17382.691489	11120.735106	49.351064	10128.819149

The products with the highest rating also have the highest average price. In contrast, the products with the lowest rating have the highest average discount.

Figure 12: Median Rating by Product Category

category	rating
Computers&Accessories Accessories&Peripherals Cables&Accessories Cables USBCables	4.1
Electronics Headphones,Earbuds&Accessories Headphones In-Ear	4.1
Electronics Home Theater,TV&Video Accessories RemoteControls	4.0
Electronics Home Theater,TV&Video Televisions SmartTelevisions	4.2
Electronics Mobiles&Accessories Smartphones&BasicMobiles Smartphones	4.1
Electronics Wearable Technology SmartWatches	4.1

Smart televisions have the highest median rating while remote controls have the lowest median rating among all categories. It is important to note, however, that due to the compression of the dataset earlier, all selected product categories are reporting a median rating of 4 or higher.

Data Distributions

Figure 13: Histogram of Product Price

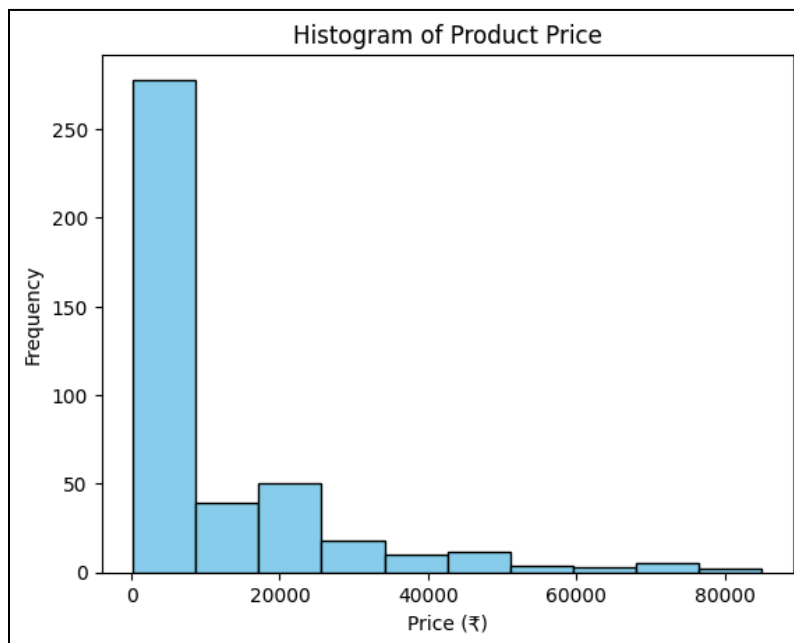
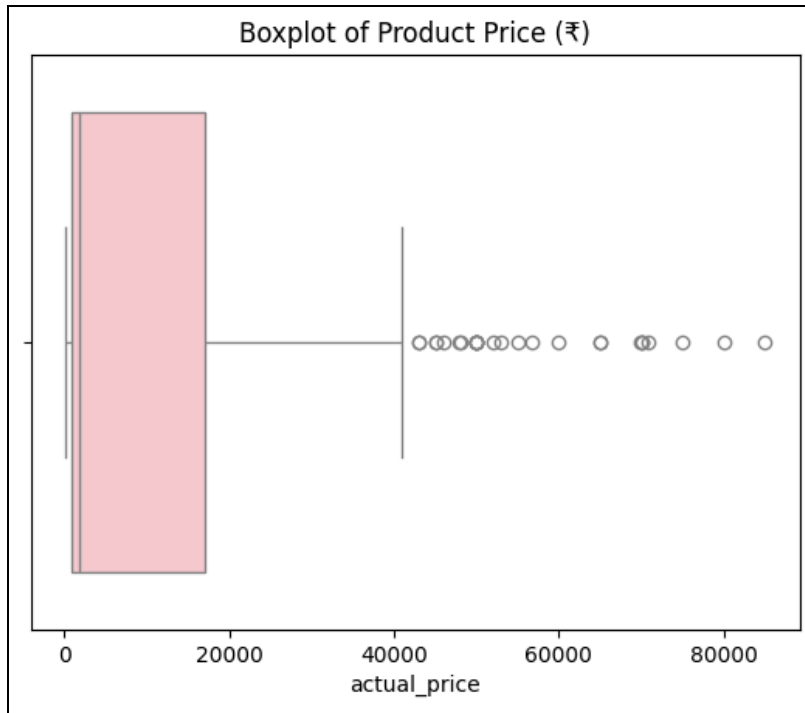


Figure 14: Boxplot of Product Price



The product price appears to be positively skewed due to many extreme data points.

Figure 15: Histogram of Discounted Product Price

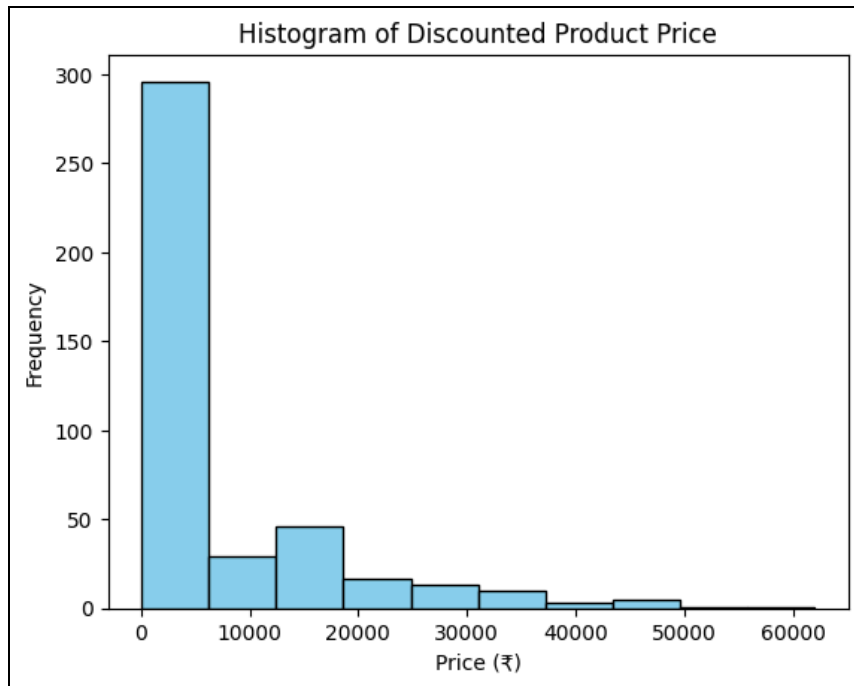
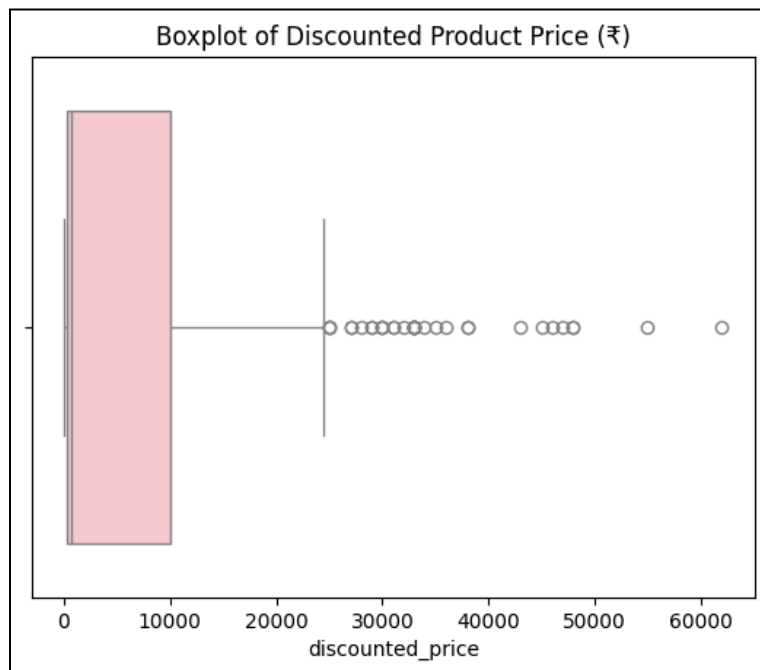


Figure 16: Boxplot of Discounted Product Price



Similarly to the distribution of product price, the discounted price is also positively skewed.

Figure 17: Histogram of Product Discount Percentage

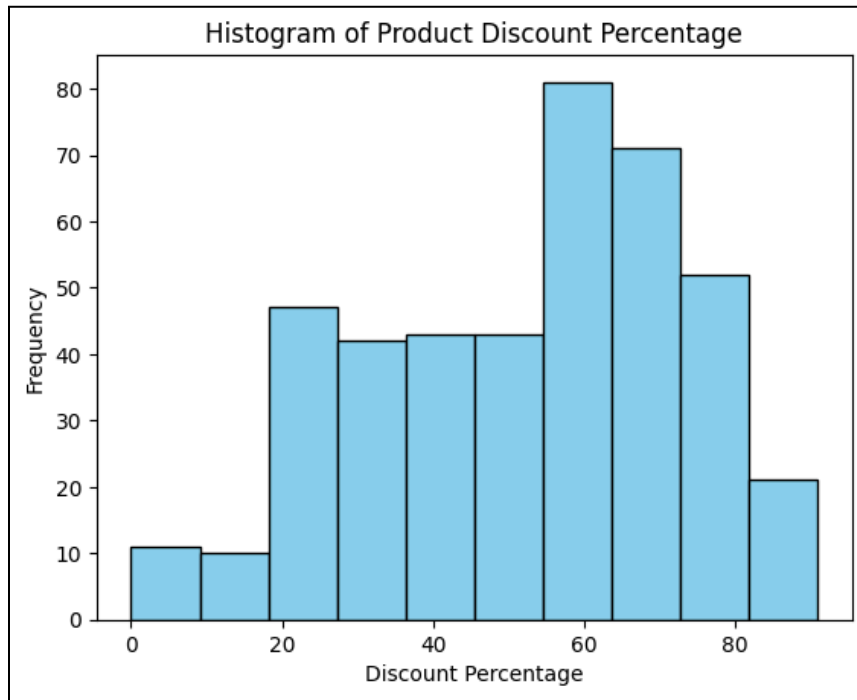
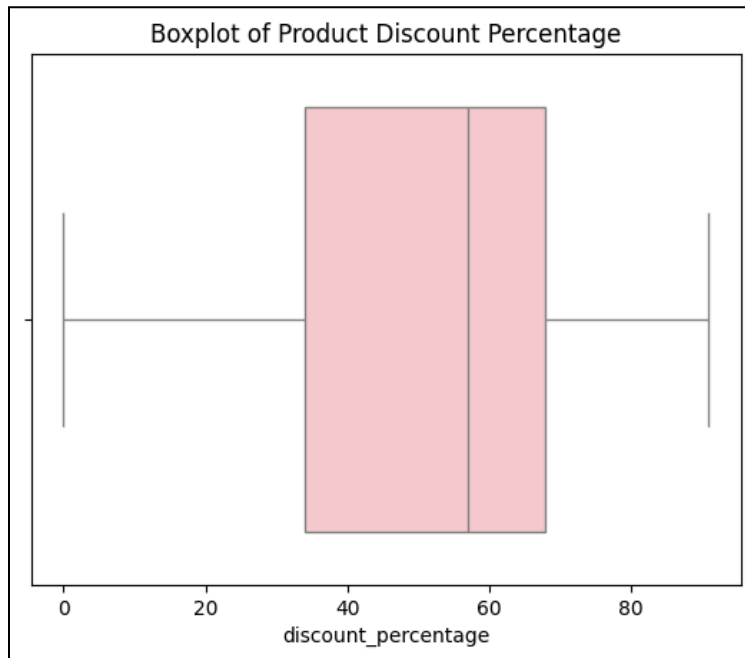


Figure 18: Boxplot of Product Discount Percentage



The distribution of the product discount percentage has a slight negative skew but is otherwise symmetric.

Figure 19: Histogram of Product Rating Votes

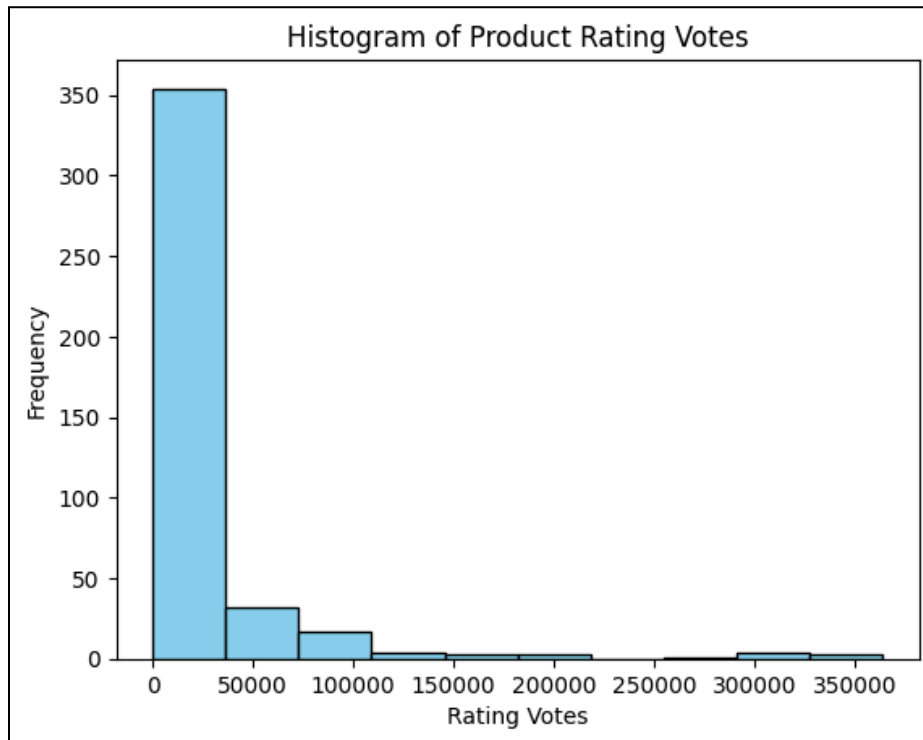
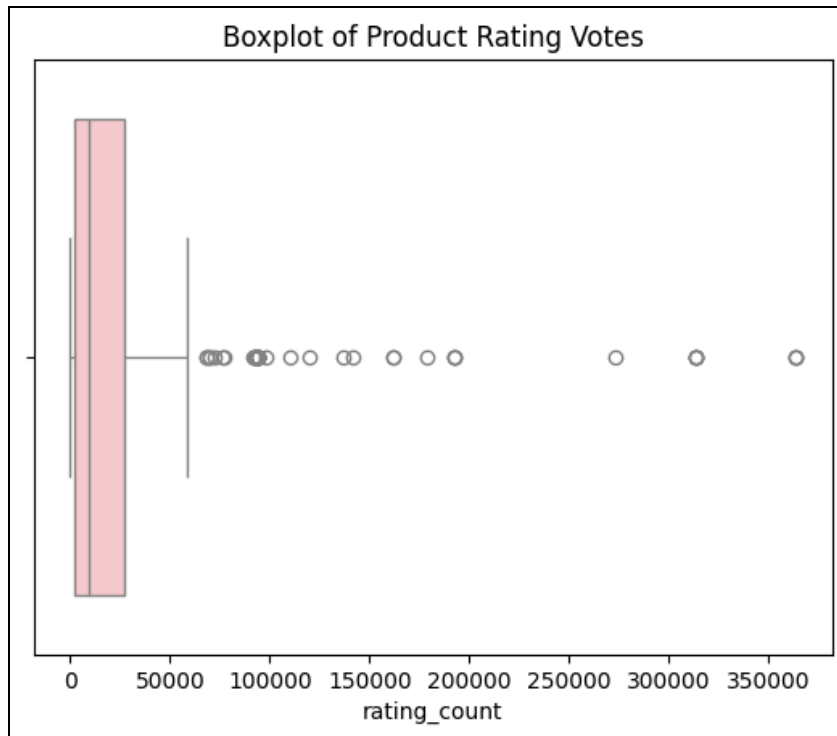
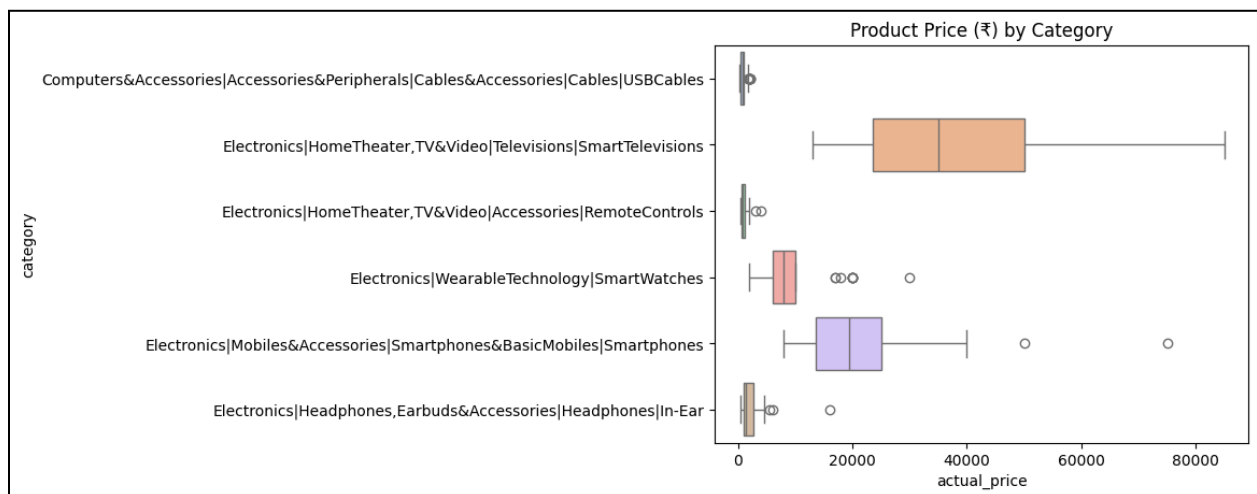


Figure 20: Boxplot of Product Rating Votes



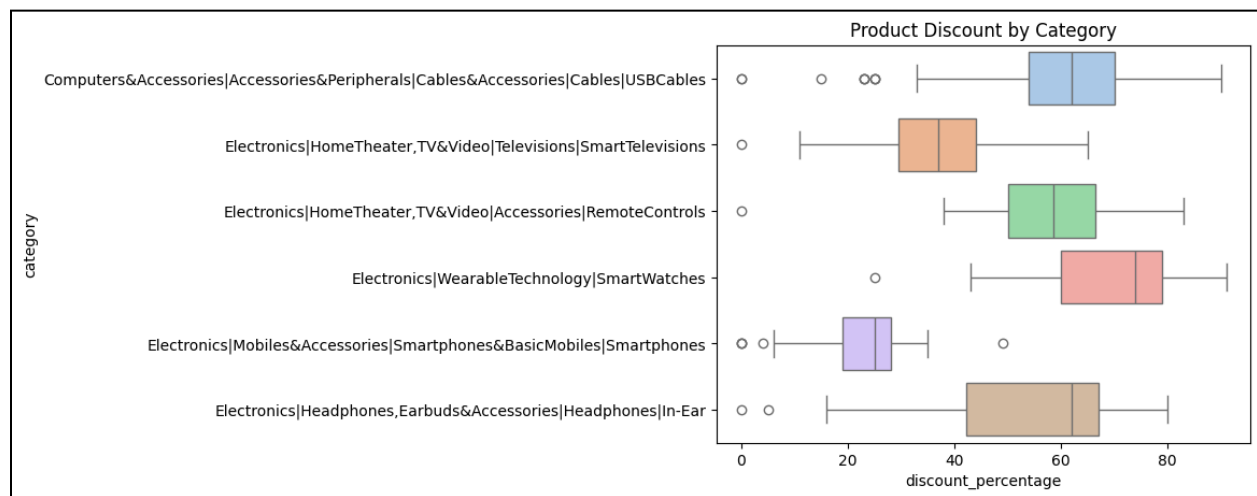
The distribution of the product rating votes has a strong positive skew, again indicating the presence of outliers.

Figure 21: Distribution of Product Price by Product Category



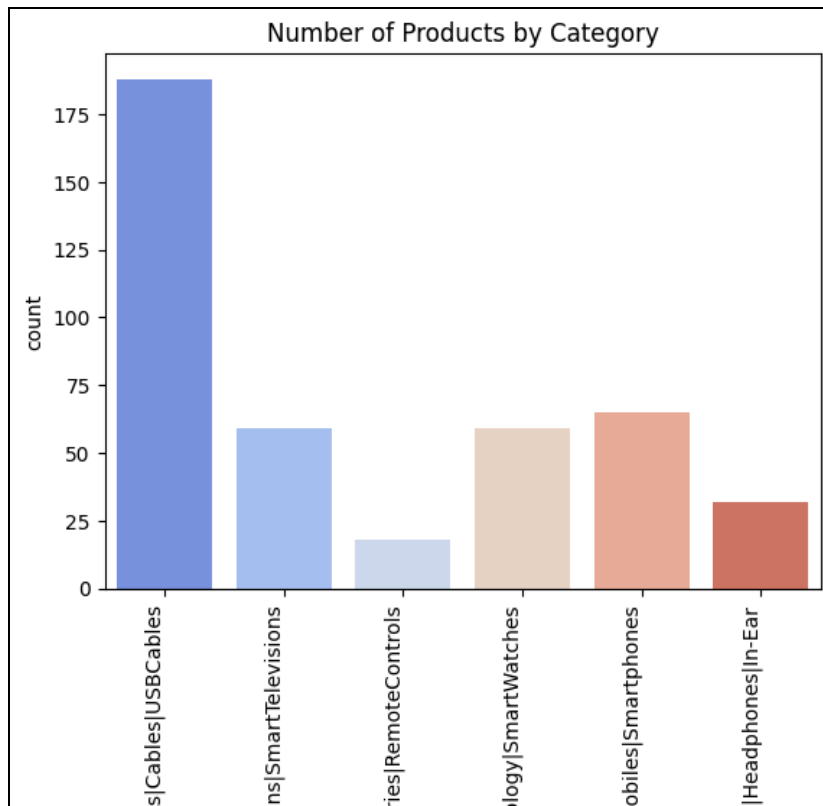
The prices of smart televisions and smartphones appear to be nearly symmetrically distributed with the fewest outliers among all categories. These two categories also have the highest product prices.

Figure 22: Distribution of Product Discount Percentage by Product Category



Smart watches have the highest offered discount while smartphones have the lowest.

Figure 23: Distribution of Product Category



The majority of the products in the dataset are USB cables while there is very little data on remote controls.

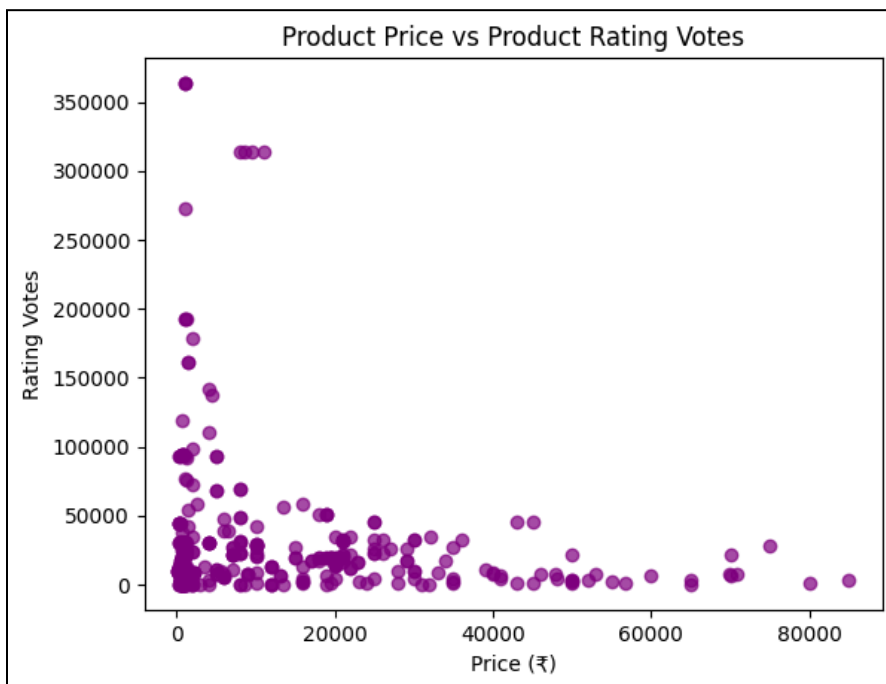
Detection of Linear Relationships

Figure 24: Relationship Between Actual Product Price and Discounted Product Price



There appears to be a strong positive linear relationship between the actual and discounted product prices which would invalidate the assumption of multicollinearity if both variables are used as predictors in a regression model.

Figure 25: Relationship Between Product Price and Product Rating Votes



There appears to be a weak negative relationship between the product price and rating votes.

Figure 26: Relationship Between Discounted Product Price and Product Discount Percentage

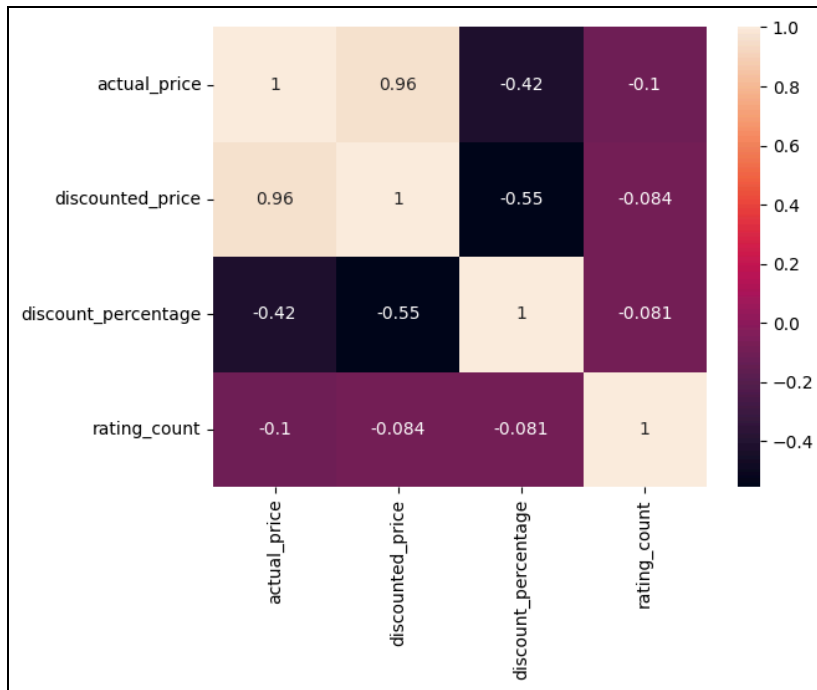


The plot suggests a potential moderate negative linear relationship between the discounted product price and the discount being offered.

Figure 27: Correlation Matrix

	actual_price	discounted_price	discount_percentage	rating_count
actual_price	1.000000	0.962910	-0.415841	-0.100234
discounted_price	0.962910	1.000000	-0.554204	-0.084120
discount_percentage	-0.415841	-0.554204	1.000000	-0.080742
rating_count	-0.100234	-0.084120	-0.080742	1.000000

Figure 28: Correlation Heatmap



Product price and discounted product price have a high correlation coefficient of about 0.96 indicating a strong potential linear relationship. A correlation of -0.42 suggests a moderate negative linear relationship between price and discount percentage while a correlation of -0.55 also implies the same between discounted price and discount percentage.

Data Preparation

The dataset contains no missing values and has been prepared for the analysis by cleaning the numerical data of any unwanted characters. Not only is the dataset large, but it also contains a high level of class imbalance for the product category and product rating variables. Both of these variables have several data points which correspond to fewer than 30 products. For this reason and as previously discussed, the data has been filtered to include only those categories and ratings which have more than 30 unique products. As we are removing data points with very few observations, the resulting dataset is considered to be representative of the entire population of products.

Data Transformation

Classification

One-hot encoding will be applied to the categorical variables to preserve semantic meaning in the analysis. The preliminary analysis uncovered three potential issues which need to be addressed prior to the model development phase: skewed distributions, outliers and

multicollinearity. Decision trees for classification purposes are generally robust to outliers in contrast to KNN and logistic regression algorithms.

Due to the large volume and extremity of outliers, the treatment of outliers as missing values by applying imputation (TOMI) technique will be used. Outliers will be replaced with the respective medians for each of the numerical variables. Furthermore, a distance-based model like KNN is highly sensitive to outliers and consequently, skewed data. Min-max scaling will be applied to ensure all numerical features in the KNN model range between 0 and 1.

Z-score standardization (normalization) will be applied to all the numerical features to maintain interpretability and for fair comparison between regression coefficients. Multicollinearity in regression can increase the variance of the model coefficients which leads to a reduction in statistical power. We previously identified potential linear relationships of concern between some of the variables. However, this information alone is not sufficient for feature selection.

After constructing a baseline multinomial logistic regression model, the VIF will be iteratively calculated to remove redundant variables. The odds ratio for each predictor will also be calculated to maximize the model's predictive power. Finally, if there appears to be no linearity between the log odds of the target and predictors, an appropriate transformation such as taking the logarithm or square root will be applied to conform to the model assumptions of logistic regression.

Information Retrieval

For the vector space models, the product descriptions will undergo tokenization prior to the application of standard text preprocessing techniques. For the baseline model, these include: lowercasing, lemmatization, stopwords removal, stemming and removal of all non-alphabetic characters.

Model Development

Model Inputs

All three of the classification models will use the following features to predict the product category target variable.

- actual_price
- discounted_price
- discount_percentage
- rating
- Rating_count

The vector space models will require only the *about_product* feature and a new product description as inputs to output a ranked list of similar products from the data.

Train-Test Split for Classification Models

The baseline models will be developed using an 80/20 data split which will reserve 80% of the data for model training and 20% of the data for testing purposes. To prevent data leakage, the mathematical transformations discussed prior will be computed using only the training data.

K-fold cross validation will be used to understand how the models generalize to unseen data and for hyperparameter tuning. After filtering the data, the remaining dataset retains only 30% of its original size. This shrink justifies the use of a 10-fold training strategy to avoid overfitting while capturing sufficient product data patterns.

Multinomial Logistic Regression

One-hot encoding is applied to the *rating* feature and the numerical features have been normalized.

Verification of Assumptions

1. The product characteristics are not dependent on one another implying independent observations.
2. The product categories are mutually exclusive.
3. While the preliminary analysis suggested potential multicollinearity between the predictors, this will be further investigated when refining the model.
4. The linearity between the log-odds of the target and the predictors will be assessed as part of the next steps.
5. Outliers have been imputed prior to building the model.

Model Output

Figure 29: Multinomial Logistic Regression Model Coefficients

	actual_price	discounted_price	discount_percentage	rating_3.9	rating_4.0	rating_4.1	rating_4.2	rating_4.3	rating_count
Cables USBCables	-3.839253	-2.019376	1.039688	0.343665	-0.432881	0.374817	-0.200167	0.079738	0.172487
Headphones In-Ear	-0.474158	-0.500091	-0.071931	0.772091	0.794809	-0.537943	1.047642	-0.038661	-1.258027
Accessories RemoteControls	-0.595886	-0.484847	0.678324	-2.050750	0.367412	0.356988	-1.237202	0.737486	-0.243184
Televisions SmartTelevisions	1.556468	1.231851	-0.737942	-0.074337	-0.657574	-0.789885	-0.568418	0.678032	1.332212
Smartphones&BasicMobiles Smartphones	1.817254	1.438559	-2.524074	0.054922	0.200811	0.197002	0.518561	-0.861373	-0.025846
WearableTechnology SmartWatches	1.535574	0.333904	1.615934	0.954408	-0.272577	0.399020	0.439585	-0.595222	0.022358

The implementation used to build the model does not assign a baseline category class which changes the interpretation of the model coefficients. Since the numerical features have been standardized, a one standard deviation increase in the product price increases the odds of the

product belonging to the smart televisions category by a factor of $e^{1.56} \approx 4.76$ compared to all other categories on average, assuming all other predictors are held constant. To understand the most influential predictors of the product category, the statistical significance of the model coefficients will be examined as part of the next steps.

Model Evaluation

Figure 30: Multinomial Logistic Regression Classification Report

	precision	recall	f1-score	support
Cables USBCables	0.91	0.91	0.91	46
Headphones In-Ear	0.25	0.20	0.22	5
Accessories RemoteControls	nan	0.00	0.00	4
Televisions SmartTelevisions	0.77	0.91	0.83	11
Smartphones&BasicMobiles Smartphones	0.56	0.71	0.62	7
WearableTechnology SmartWatches	0.77	0.83	0.80	12
accuracy			0.80	85
macro avg	0.65	0.59	0.57	85
weighted avg	0.80	0.80	0.78	85

The model is able to predict the product category correctly 80% of the time. As seen by the *support* column, despite filtering the data earlier, there is still a significant class imbalance in the testing data. This suggests that the train-test split should be modified to construct a more representative testing dataset. This class imbalance supports the use of the weighted average precision metric. A precision of 80% indicates that out of all the times the model predicted a product belonging to a particular category, 80% of those products were accurately classified. Similarly for recall, 80% of the time the model was able to identify products actually belonging to a particular category. For instance, for all products belonging to the smart televisions category, the model identified 91% of these products correctly. The majority of these scores are below 80% which suggests the need for model improvement.

KNN

One-hot encoding is applied to the *rating* feature and the numerical features have been scaled using min-max scaling.

Model Evaluation

Figure 30: KNN Classification Report

	precision	recall	f1-score	support
Cables USBCables	0.87	0.89	0.88	46
Headphones In-Ear	0.50	0.60	0.55	5
Accessories RemoteControls	0.25	0.25	0.25	4
Televisions SmartTelevisions	0.62	0.73	0.67	11
Smartphones&BasicMobiles Smartphones	0.57	0.57	0.57	7
WearableTechnology SmartWatches	1.00	0.67	0.80	12
accuracy			0.76	85
macro avg	0.63	0.62	0.62	85
weighted avg	0.78	0.76	0.77	85

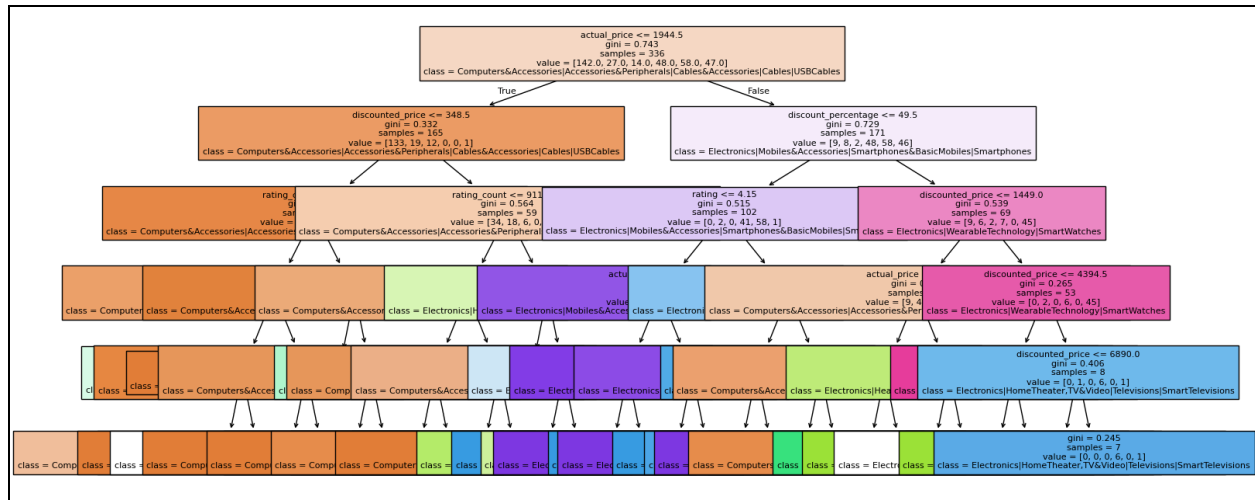
By default, the model predicts the target class using the 5 nearest neighbours. The best choice for this hyperparameter will be determined later. The overall scores for the KNN model appear to be lower than the previously built regression model. The model is able to correctly classify the product category 76% percent of the time. Out of all the category predictions made by the model, 78% of those predictions reflect the true product category. For all products belonging to a particular category, the model is able to identify these products 76% of the time. For instance, 89% of all USB cables are correctly retrieved by the model.

Decision Tree

Note that one-hot encoding is not required for the categorical *rating* feature because the data is already label-encoded and ordinal by nature.

Model Output

Figure 31: Decision Tree Visualization



The product prices features appear to be the most important with the lowest Gini impurity when initially constructing the tree.

Model Evaluation

Figure 32: Decision Tree Classification Report

	precision	recall	f1-score	support
Cables USBCables	0.94	1.00	0.97	46
Headphones In-Ear	1.00	0.60	0.75	5
Accessories RemoteControls	1.00	0.25	0.40	4
Televisions SmartTelevisions	0.77	0.91	0.83	11
Smartphones&BasicMobiles Smartphones	0.83	0.71	0.77	7
WearableTechnology SmartWatches	0.85	0.92	0.88	12
accuracy			0.89	85
macro avg	0.90	0.73	0.77	85
weighted avg	0.90	0.89	0.88	85

The gini index is used as the split criterion and the *max_depth* parameter is set to 5. Further hyperparameter tuning will be performed in the next stage of the analysis. The decision tree model has reported the highest scores compared to all other models, with an ability to accurately identify the product category 89% of the time. From all the predictions the model makes for a particular class, 90% of the time the predictions correspond to the actual product category. For instance, when the model predicts a product to be a smart watch, 85% of the time it is correct. In comparison, the model is able to correctly classify 92% of all smart watches.

Information Retrieval Pipeline

The product descriptions have undergone extensive text-preprocessing.

- Lowercase all letters
- Replace digits with the string *num*
- Tokenize the text
- Remove non-alphabetic tokens
- Remove stopwords
- Lemmatization
- Stemming

The TF-IDF statistical weight measure was used to identify relevant terms from the product descriptions in a unigram vector space model.

Model Evaluation

Figure 33: Cosine Similarity Report

index	ID	Category	Description	Cosine Similarity
0	Doc401	Electronics WearableTechnology SmartWatches	1.75" TruView™ display: See the clear, bigger picture on the 1.75" touch screen with 320*385 pixels Grade 6061 aluminium body: Crafted from Grade 6061 aluminium, ColorFit Ultra can withstand daily rough use with ease. Personal health assistant: Take care of your health with features like blood oxygen, heart rate, stress, REM & sleep monitors SpO2 monitor: Measure your blood oxygen level anytime on the SpO2 monitor, with just a tap. 60 sports modes: With 60 sports modes, you are spoilt for choices; just pick and play your sport.	1.0
1	Doc319	Electronics WearableTechnology SmartWatches	1.75" TruView™ display : See the clear, bigger picture on the 1.75" touch screen with 320*385 pixels Crafted from Grade 6061 aluminium, ColorFit Ultra can withstand daily rough use with ease.Strap Size & Material : 22mm & Silicone Personal health assistant & NoiseFit App : Take care of your health with features like blood oxygen (SpO2), heart rate, stress, REM & sleep monitors. Know more about your health, stay connected with the world and more via the dedicated NoiseFit app. 60 sports modes & 100+ watch faces: With 60 sports modes, you are spoilt for choices; just pick and play your sport. Change your look as per your preference with 100+ customisable and cloud-based watch faces (and straps). 9-day battery & IP68 Waterproof : The powerful 9-day battery stays with you through your everyday hustle. ColorFit Ultra is ready to go with you everywhere, thanks to IP68 waterproof rating :. Productivity features : With smart utility features like call notification, quick reply SMS, world clock, stock market info, torch and more, you will be on top of your game. Charging Time : Upto 2 hours Item Type Name: Smartwatch	0.6932729827969166
2	Doc410	Electronics WearableTechnology SmartWatches	1.55 Always-on AMOLED Display with high 314 PPI Resolution for a crystal clear image and sharp details Super-Light weight body with only 19.5 grams, 8.95 MM Thin body, comfortable to wear day & night and during sports activities. Monitoring your blood oxygen levels by intelligently monitoring your (SpO2) blood oxygen levels. Basic Usage Battery Life: 21 days 68 built-in sports modes, 5 ATM water-resistance and Alexa Built-in 24H Heart Rate Tracking, Female cycle tracking, Sleep quality monitoring and Stress level monitoring Bluetooth music control directly from your watch while exercise without having to pull out your phone. Check Out Our Store : Click on the blue Amazon link below the Product title, to explore our other models.	0.496243554762996
3	Doc371	Electronics WearableTechnology SmartWatches	[10-day battery]: With a battery backup of 10 days, just concentrate on your game without stopping. [Noise Health Suite]: Automatically monitor your blood oxygen (SpO2 levels), real-time heart rate, and provide comprehensive analysis of your sleep quality (deep sleep, light sleep and wake up time) on this smart watch. [Personalise your look]: Choose from 60+ cloud-based watch faces or customise them as per your preference. [1.4" Full Touch HD display]: Get the perfect viewing experience on the 1.4" display with 240*240 pixels. 8 sports modes: Choose from 8 sports modes and give it your best.	0.4678152552372992
4	Doc274	Electronics WearableTechnology SmartWatches	[10-day battery]: With a battery backup of 10 days, just concentrate on your game without stopping. [Noise Health Suite]: Automatically monitor your blood oxygen (SpO2 levels), real-time heart rate, and provide comprehensive analysis of your sleep quality (deep sleep, light sleep and wake up time) on this smart watch. [Personalise your look]: Choose from 60+ cloud-based watch faces or customise them as per your preference. [1.4" Full Touch HD display]: Get the perfect viewing experience on the 1.4" display with 240*240 pixels. 8 sports modes: Choose from 8 sports modes and give it your best.	0.4678152552372992

Figure 33 displays the top 5 products from the dataset which are the closest match to the product with an ID of *Doc401*. Cosine similarity is calculated by measuring the cosine of the angle between the search query vector and each of the vectors in the TF-IDF matrix. The similarity scores range between 0 and 1, with a higher score indicating a closer match. The model reports a perfect cosine similarity with *Doc401* as it is identical to the product of interest. *Doc319* is the next closest match with a cosine similarity score of about 0.69. Note that the top 5 product matches all belong to the smart watches category which suggests acceptable model performance.

Next Steps

The models presented in this analysis are purely baseline and require further refinement. Outlined below are some of the key improvements which will be considered during the next stage.

- Modify train-test split to account for class imbalances
- Tune hyperparameters for the classification models using 10-fold cross validation
- Address multicollinearity and verify the assumption of linearity in the regression model
- Construct bigram vector space models and evaluate search relevance

External Files

All work pertaining to this project can be accessed via the following GitHub repository:
<https://github.com/harman-khehara/Amazon-Product-Analysis>.

GenAI Disclosure

The entirety of this project will be completed without the use of any generative artificial intelligence. Development history of the project will be tracked using GitHub and Google Suite. Justification of all work can be provided upon request.

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